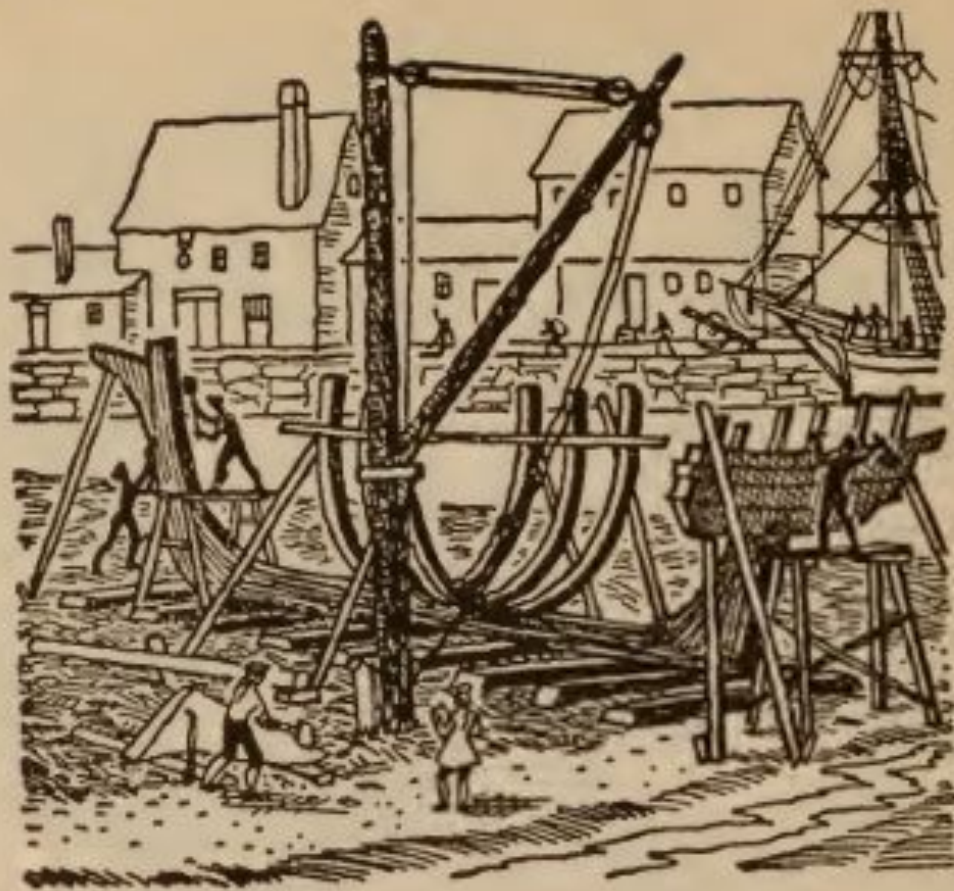


Carry On, Mr. Bowditch

Chapter 7



CHAPTER 7



THE ALMANAC

Before winter came again, Nat could find anything in the shop as quickly as Mr. Ropes or Mr. Hodges. But Sam still dropped in to talk. Sometimes Dr. Bentley dropped in, too.

Ben Meeker was lounging in the chandlery one day when Dr. Bentley stopped by. When the roly-poly minister had gone, Ben said, "Bright fellow. Powerful bright. Knew twenty languages when he was only twenty-five. No brighter than you'd have been, though, if you weren't becalmed."

The rest of that day, Nat found it hard to forget Ben Meeker.

Most of the time, though, he did not waste much time thinking of Ben's words. All day, he was too busy in the chandlery, or running errands. At night, he was too busy writing down everything he had learned that day.

One day an errand took him to a long building called a ropewalk. He watched the ropemakers walking backwards as they twisted the fibers into yarn, the yarn into strands, and the strands into rope. The ropemakers were proud of their work. "Most important thing on a ship," they said. "You can't sail a ship without cordage!"

That night Nat started a notebook on everything about rope.

Once an errand took him to a sail loft. He watched the sail makers cutting heavy canvas into strips and sewing it back together again. "Makes the sail stronger," they told him. "Most important thing on a ship — the sails. If it weren't for sails, how'd you get anywhere?"

Nat added more to his notebook; everything about sails.

Another errand took him to Ruck's Creek, called "Knocker's Hole," because all day you could hear the *thump, thump, thump* of the caulker's mallets, caulking the seams of ships. "Most important work on a ship," they told him. "If a ship isn't watertight, where'd you be?"

Nat added more notes — this time about caulking a ship. When Sam dropped in again, Nat showed him his new notebook. “Is there anything else to know about ships?”

Sam chuckled. “Lad, you haven’t even begun! Navigation — that’s something else again! Want to learn it? I reckon I could teach you.”

So the winter Nat was thirteen, he started a new notebook: NAVIGATION Nathaniel Bowditch His Book.

By spring, the notebook was filled. Sam looked

through it one day. "It's all down there, lad. Get it in your head, and you'll have it!"

"What else can I learn?"

Sam shook his head and chuckled. "Blessed if I know! You've lightered all the cargo in my head!"

Mr. Ropes strolled back to Nat's desk. "Nat, run over to my house, and look up Surveying in the Chambers *Cyclopaedia*, will you? Hetty will show you where it is. Write down what it says about the start of surveying. You'll find everything you need on the desk."

Nat hurried over to Mr. Ropes' home. A cyclopaedia? What in the world was that? Well, Hetty would show him. . . .

Soon he was sitting at the desk in the library, with four big books in front of him: Ephraim Chambers' *Cyclopaedia, or Universal Dictionary of Arts and Sciences*. He turned the pages. Everything was here! Everything! I'd like to begin at A, he thought, and read right through to Z! But now he must find out about surveying.

The next thing Nat knew, Mr. Ropes was striding into the library, calling, "Nat! What in the name of sense happened to you? Did you go to sleep?"

"No, sir. I'm copying what it says about surveying. But there's a good bit to look up. It's a little hard to tell where the start of it is. I've looked up trigonometry — that's the kind of mathematics they use — and I've looked up theodolites — that's the kind of telescope they use. Then there's something about finding your position by sighting a star, so I got into astronomy. I can't tell yet where surveying starts — with astronomy, or trigonometry or the theodolite, or — "

"Nat Bowditch!" Mr. Ropes threw himself in a chair

and laughed until he wiped his eyes. "Where did it start? In what country? That's all I wanted to know! Give me that book a minute! . . . See? Right here! It says that surveying probably began in Egypt! So I was right!"

Nat said, "Oh . . . that's all you wanted to know?"

Mr. Ropes was still chuckling. "That was all. I'm sorry I put you to all that work for nothing."

Nat grinned. "It wasn't work. It was fun. May I keep the notes I made?"

"Of course. If you want to look up some more things in Chambers, help yourself any time. But" — he smiled — "after the chandlery closes, eh? Not in the middle of the afternoon." Mr. Ropes got up, still chuckling. "You'd better run along now. It's suppertime."

"What!" Nat jumped up and stared out the window. "Why — I — I've been here all afternoon!"

"You certainly have."

"I'm sorry. I'll finish up my work tonight."

Mr. Ropes said, "No need to do that."

But after supper Nat went back to the chandlery to finish the work he had not done. It was an unexpectedly warm night for early spring. He left the upper half of the Dutch door open.

He was busy at work when a voice called, "Nat?"

He looked up. Lizza was standing in the door. He smiled and hurried to let her in.

Lizza hesitated. "It's all right to stop a minute to talk to you? Mary always says we mustn't tag in here and bother you when you're busy."

"It's all right, Lizza."

They talked a while; then silence fell. "Funny — I had so much to tell you," Lizza said. "Now I can't seem to think of it. . . . I miss you, Nat. I . . . I . . . When I go to tell you a joke and you're not there . . . I miss you most awfully." Suddenly Lizza wheeled and ran out the door. Nat knew she was crying.

His throat ached. "A good thing Hab taught me that boys don't blubber," he muttered. He went back to his work. After he'd finished his bookkeeping, he started a new notebook: THE PRACTICAL SURVEYOR: Nathaniel Bowditch. County of Essex and State of Massachusetts, New England. March the Seventh, 1787.

When Sam saw the new notebook, he said, "Now that's something I can help you with, too. Surveying's a lot like navigation — only it's a heap easier."

"You're sure it's easier?"

Sam chuckled. "Yes, sir! When you set up your theodolite good and level with your plumb line, the ground holds still. It won't heave and pitch like the deck of a ship when you're trying to shoot the sun. Yes, sir, surveying's lots easier. You'll get along fine with it."

Nat was still working on surveying when he got his first glimpse at an algebra book. That night, for the first time, he studied all night. The sky was paling in the east when he started another notebook: ALGEBRA AND MATHEMATICS: Nathaniel Bowditch His Book.

Every time he had a chance he borrowed the algebra book, to copy it into his notebook. Between times, he copied everything on mathematics he could find in the *Cyclopaedia*. Then he studied everything he could find on astronomy over again.

He was sixteen the summer he figured how to make an almanac. He felt a tingle go up his backbone. Just to

think! A man could sit right here and figure out when the moon would rise every night next month — or next year — or ten years from now! He could figure out the way the sun would act; he could figure . . .

Ben Meeker shuffled into the chandlery one day. "What's that you're figuring on?"

"An almanac for the years from 1789 to 1823."

Ben sniffed. "Do tell. And what's your almanac going to have in it?"

"Just the regular things: the sun's rising, setting, declination, amplitude, place in the ecliptic — "

Ben stiffened. "You've no need to make fun of me!"

Nat stared. "But I'm not making fun. You asked me what is in my almanac, and I was telling you."

A strange voice said, "Pardon me — may I see your almanac?"

"Of course, sir." Nat handed his almanac to the stranger, and turned back to Ben. "It's just straight mathematics, Ben. You see —"

Ben threw up his hands. "Don't tell me! Save it for them as had a chance to go to school!" He shuffled out.

Nat turned to the stranger. "Something for you, sir?"

"Yes. A compass. Just a small one, please. For a child.

We live in Cambridge. My little daughter says the streets get her mixed up."

Cambridge! Where Harvard was! Nat brought him the compass and gave him his change. At last the man looked up from the almanac.

"How old are you?"

"Sixteen, sir."

"But — but — this is amazing. You ought to be — Have you ever thought of going to Harvard?"

"I can't leave here, sir. I'm indentured."

The man frowned. "But that's — that's — " He stopped. He wrote on a slip of paper and gave it to Nat. "If anything happens that you can leave here — say within a year or two — write to me. I'd like a tutor for my children. I'm sure you could do that and go to Harvard, too. An almanac at sixteen!"

For weeks after that, Nat used to imagine the letter he'd write:

Dear Mr. Morris

I'm free now, and can come to Cambridge and be a Harvard man. The thing that freed me from my indenture was —

Nat's letter always stopped there. He couldn't think of anything that could happen to free him until he was twenty-one. And that was still five years away. Sometimes he agreed with Ben. Nine years was a long time to sail by ash breeze.